BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LANs, WANs and Network Performance

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Why is a thermostat controller an embedded system?

2. After one ascending bubble-sort pass on [5, 2, 4, 1], what is the list?
   A) [2, 4, 1, 5]
   B) [1, 2, 4, 5]
   C) [5, 2, 4, 1]
   D) [1, 4, 2, 5]

3. Insertion sort has sorted part [2, 5, 8] and next value 3. Which values move? Give the new sorted part.

4. A program tests whether score >= 50. "A comparison operator must always produce a text paragraph." Correct this. What could happen?

Today's objectives

Define computer networks and distinguish a LAN from a WAN. Explain bandwidth, number of users and media as performance factors.

Key vocabulary - Read, cover, recall

Network
Two or more connected devices sharing resources.

LAN
Local Area Network; small geographical area.

WAN
Wide Area Network; larger geographical area.

Bandwidth
Data a connection can transfer per second.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Computer networks

Two or more connected computers share data and resources, such as printers, files and an internet connection.

LAN compared with WAN

A LAN covers a small area such as a school. A WAN covers a larger area and may link multiple LANs.

Network performance

Bandwidth measures how much data can be transferred per second. Users share available capacity; wired and wireless media can have different limits.

Explain the effect

Do not claim all WANs are slower. Use the scenario to link bandwidth, demand or transmission media to the impact on performance.

I Do - Worked model

A trust links two schools in different towns. Explain why it needs a WAN and how busy connections may affect performance. [4 marks]

- Schools in different towns cover a larger geographical area.
- A WAN links geographically distant networks [1] including the two school LANs [1].
- Many simultaneous users share the available connection capacity [1], so file transfers may become slower [1].
- Two WAN points and two linked performance points give four marking points.

We Do - Guided practice

1. A library connects its 15 PCs and a printer in one building. Explain why a LAN suits it and how many users could affect performance. [4 marks]

2. A school upgrades to a connection with more bandwidth. Explain two likely benefits. [4 marks]

3. A classroom uses wireless tablets. Explain two factors that could slow the network. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the estimated transfer time in seconds.

```
fileMb = 48
rateMbps = 12
print(fileMb DIV rateMbps)
```

You Do - Independent exam questions

Q2. State what LAN and WAN stand for. [2 marks]

Q3. An online gaming event at a youth club experiences delays when many players connect. Explain two factors affecting its network performance. [4 marks]

Q4. A retail company has a network within each shop and connects shops in different cities. Compare LANs and WANs and describe two factors that could affect performance. [6 marks]

Exit Ticket - True or False?

Every WAN is always slower than every LAN.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It is part of a heating system, with the specific task of controlling temperature.

2. A - [2, 4, 1, 5]

3. Values greater than 3 shift right: [5, 8]. New prefix [2, 3, 5, 8].

4. A comparison gives a Boolean result: true or false. It compares values; assignment stores a value.

I Do - Explain the model

Schools in different towns cover a larger geographical area.

A WAN links geographically distant networks [1] including the two school LANs [1].

Many simultaneous users share the available connection capacity [1], so file transfers may become slower [1].

Two WAN points and two linked performance points give four marking points.

We Do - Model answers

1. A LAN covers a small area [1] and allows sharing the printer [1]. Many users share available bandwidth [1], which may slow transfers [1].

2. More data can be transferred per second [1], reducing transfer time for a file [1]. More simultaneous users may be supported [1] with fewer delays [1].

3. Many simultaneous users share bandwidth [1], slowing transfers [1]. Wireless interference [1] can disrupt data transmission and reduce reliability [1].

Code First - Answer

```
fileMb = 48
rateMbps = 12
print(fileMb DIV rateMbps)
```

Expected output: 4

You Do - Self-mark

Q2: Local Area Network [1]; Wide Area Network [1].

Q3: Many users share capacity [1], which may slow individual transfers [1]. Limited bandwidth [1] limits the amount of data transferred per second [1].

Q4: LAN small area [1], WAN large geographical area [1]. Bandwidth affects data transfer per second [1] and may change transfer speed [1]. Number of users affects available bandwidth per device [1], potentially causing delays [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.